FIN Research Supplement — Release 10.3

FIN Negative Information Coupling

Executed Programs 31–40, the Legacy-to-Strict Bridge, and the Operational Meaning of Information Loss

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19 July 2026

Publication — Preprint; record version 1.0.0

Scope. Ten executed mathematical and computational studies of a conditional negative-information-coupling extension. The report distinguishes static attenuation, stochastic contraction, open-system information transfer, feedback, and fundamental deletion.

Repository. `https://github.com/hyconiek/Fractal-Nadsoliton-Theory`
Guardrail. The legacy kernel is treated as an intended intermediate precursor, not as an already proven identity with the strict kernel. No role transfer, selector closure, physical unit, or Theory-of-Everything claim is made.
Reproducibility. Code, JSON results, nine figures, Markdown, LaTeX, Zenodo description, and checksums accompany the PDF.

Zenodo DOI. To be assigned to this record.

Publication statement

This preprint executes Programs 31–40 and tests negative information coupling as a new conditional bridge component. It is self-contained and reports proofs, numerical evidence, counterexamples, identifiability limits, and reproducibility artifacts. It does not claim a completed legacy–strict bridge, transfer of legacy physical roles, a strict selector, physical units, fundamental information destruction, hardware validation, or a Theory of Everything.

Contents

Confidence convention

- **[Proven]** means that an analytic proof is given or that a finite claim is reduced to auditable algebra.
- **[Strong evidence]** means that a reproducible numerical result has a wide stability margin but is not promoted to a general theorem.
- **[Moderate evidence]** means that the conclusion depends materially on the declared model class.
- **[Speculative]** means that a mathematically admissible construction lacks a source theorem or decisive test.
- **[Refuted]** means that the stated claim has an analytic obstruction or an explicit counterexample.

Every time, rate, action, temperature, and coupling used below is dimensionless. No SI unit, physical clock, physical environment, selector, or experimental interpretation is inferred from either kernel alone.

Chapter 1

Executive summary

This monograph executes ten new studies motivated by the hypothesis that the transition from the legacy FIN kernel to the strict kernel may contain an additional negative information coupling. The hypothesis is accepted only in a repository-compatible conditional form:

> *The legacy kernel is the intended intermediate foundation or precursor of the strict kernel, but the repository does not yet contain a theorem identifying the two. Negative information coupling is tested as a possible additional completion component, not assumed as a strict-derived fact.*

The first result is semantic but decisive. The historical transformation document contains spatial attenuation, oscillatory signs, and heuristic path summation. It does not define a state, time law, channel, environment, instrument, measurement record, or information functional. Spatial attenuation of a kernel is therefore not yet operational information loss. [**Proven by structural audit**]

The ten executed programs establish the following.

1. A positive retention factor cannot transform the canonical legacy row—which is positive at distances 1 and 6 but negative at distances 2 through 5—into the positive six-distance strict row. Negative information loss can contribute an envelope, but a separate phase/frequency or distance-mixing map is necessary. **[Proven]**
2. On the canonical twelve-cycle, the finite operator difference between the strict and legacy Laplacians is positive semidefinite to numerical precision; on an interval extrapolation of the same analytic formulas it has a negative eigenvalue, approximately minus 4.69964. The interpretation "strict equals legacy plus passive dissipation" is geometry-dependent. **[Strong evidence]**
3. The raw signed legacy kernel is not a Markov generator. Its candidate Laplacian has a minimum eigenvalue near minus 21.3771, and its exponential at dimensionless time 0.001 already contains negative entries. Two elementary positivity repairs are admissible but have spectral gaps differing by a factor of 21.37. **[Proven]**
4. The change from the linear legacy envelope to the strict power-1.8 envelope has an exact positive discrete-hazard representation. That representation is a reparameterization, not a microscopic derivation of information loss or of the exponent 1.8. **[Proven]**
5. Exponential retention preserves the graph-Laplacian cone for every nonnegative feedback strength. Naive linear subtraction ceases to define Markov rates once the feedback strength exceeds one sixth. **[Proven]**
6. Apparent system information loss is exactly compatible with transfer to an environment. At dimensionless time one, the strict Markov channel reduces relative information to 0.258910

nats, while an explicit isometric dilation keeps the joint state pure and stores 4.451992 nats of system–environment mutual information. **[Proven]**

7. The sign of a system–environment coupling is an exact system-only gauge for a parity-symmetric bath. A polarized bath or calibrated odd environment record restores sign sensitivity. **[Proven]**
8. Stable negative feedback need not be variational. The tested two-coordinate feedback has eigenvalues with real part minus 0.8 and imaginary parts plus or minus 1.7, a normalized integrability defect of 1.80964, and nonzero closed-loop work. [**Proven for the declared model**]
9. The same feedback-modified spatial operator still supports wave and diffusion categories. Their early escape exponents remain approximately two and one, respectively, and only diffusion satisfies Chapman–Kolmogorov composition at the population level. Negative coupling does not resolve the observer or temporal-category ambiguity. [**Proven analytically; numerically verified**]
10. In the synthetic held-out challenge, four declared models were classified with 85.4 percent accuracy. The negative model was identified in 93.3 percent of its trials. However, its system-only signature was exactly identical to a static mimic; only calibrated environment records separated the causal stories. **[Moderate evidence]**

The deepest surviving conclusion is:

> *Negative information coupling can be made into a coherent operational addition to FIN, but it is not hidden in the static legacy formula and it cannot complete the legacy-to-strict bridge by itself. The smallest admissible completion needs at least a phase/frequency transformation, a positive retention or dissipative map, a state space, a temporal category, an environment or coarse-graining map, an information functional, and an operational sign reference.*

No legacy physical role is transferred to the strict kernel in this work.

Chapter 2

Part I — Mathematical input and scope

2.1 The two kernel objects

The canonical legacy object is

$$K_\ell(d) = \alpha_{\text{geo}} \frac{\cos(\omega_\ell d + \phi_\ell)}{1 + \beta_{\text{tors}} d},$$

with

$$\alpha_{\text{geo}} = 4 \log 2, \qquad \omega_\ell = \frac{\pi}{4}, \qquad \phi_\ell = \frac{\pi}{6}, \qquad \beta_{\text{tors}} = 0.01.$$

The later strict working object is

$$K_s(d) = \frac{\cos(\omega_s d + \phi_s)}{1 + \beta_s d^{\eta_s}},$$

with

$$\omega_s = 0.18575, \qquad \phi_s = 0.16250, \qquad \beta_s = 1, \qquad \eta_s = 1.8.$$

The strict six-distance row is

$$(0.469985673, 0.192043552, 0.091428614, 0.047029169, 0.024131223, 0.011070817).$$

The project guardrails identify K_ℓ as an intermediate legacy kernel and K_s as a completed or enriched strict working kernel only where an explicit completion certificate licenses the passage. The actual strict derivation chain is an operational refreeze and gate-selection chain, not a finished derivation from K_ℓ. This report therefore tests a candidate bridge without silently assuming it.

2.2 What "negative information coupling" may mean

Four notions must not be conflated.

2.2.1 Negative kernel entry

$$K_{xy} < 0.$$

This may denote phase, anticorrelation, or a signed interaction. A Hermitian matrix with negative entries can generate perfectly unitary dynamics and no information loss. **[Proven]**

2.2.2 Static attenuation

$$K'(d) = r(d)K(d), \qquad 0 \le r(d) \le 1.$$

This suppresses a spatial coefficient. Without a state and time law it is not an entropy statement. **[Proven]**

2.2.3 Classical operational loss

For a Markov semigroup $P_t = e^{-tA}$ with stationary uniform state u, define

$$\mathcal{I}(p) = D(p\|u) = \log 12 - H(p) \ge 0.$$

If P_t is doubly stochastic, data processing gives

$$\mathcal{I}(P_t p) \le \mathcal{I}(p).$$

Here "loss" means reduced distinguishability relative to a declared accessible state space.

2.2.4 Quantum local loss

A minimal dephasing completion using a self-adjoint operator A is

$$\dot{\rho} = -i[A,\rho] - \frac{\gamma}{2}[A,[A,\rho]], \qquad \gamma \ge 0.$$

In the eigenbasis of A,

$$\rho_{jk}(t) = e^{-it(a_j - a_k)} e^{-\gamma t (a_j - a_k)^2/2} \rho_{jk}(0),$$

and

$$\frac{d}{dt} \operatorname{Tr} \rho^2 = -\gamma \|[A,\rho]\|_{\mathrm{HS}}^2 \le 0.$$

This is a legitimate completely positive open-system law in the framework of the GKS theorem and Lindblad theorem. It requires an additional dephasing rate, a time variable, a state space, and an open-system interpretation. None is selected by a static kernel. **[Proven]**

2.3 Arithmetic audit of the historical transformation diagram

DIAGRAMS_KERNEL_TRANSFORMATION.md presents the historical route

$$K_{\text{total}} = K_{\text{geo}} K_{\text{res}}(1 + 0.2K_{\text{tors}})K_{\text{topo}} \rightsquigarrow K_\ell(d).$$

It is valuable as a record of the intended mechanism but contains three exact arithmetic discrepancies.

First,

$$e^{-2.9\cdot 7} = 1.52694 \times 10^{-9}, \qquad e^{-2.9\cdot 12} = 7.70109 \times 10^{-16},$$

not 9×10^{-4} and 6×10^{-6}.

Second,

$$d^{1.6}d^{-0.6} = d^{+1},$$

not d^{-1}. With $d^{1.6}$ paths, an inverse-distance tail would instead require a mean amplitude of order $d^{-2.6}$.

Third,

$$\cos\left(\frac{\pi}{4}d + \frac{\pi}{6}\right) = 0 \quad \iff \quad d = \frac{4}{3} + 4n,$$

so the stated integer node sequence is not exact.

These observations do not invalidate the frozen legacy formula. They prove that the diagram is not itself a theorem-level derivation of the legacy kernel or of an operational loss mechanism. **[Proven]**

2.4 Minimal operational completion tuple

To speak rigorously about information loss, the kernel must be embedded into at least

$$\mathfrak{O} = (\mathcal{S}, A, \mathcal{T}_t, \mathcal{I}, \mathcal{E}, \mathcal{J}, \mathcal{R}),$$

where:

- $\mathcal{S}$ is a state space;
- A is a generator or Hamiltonian;
- $\mathcal{T}_t$ is the temporal evolution;
- $\mathcal{I}$ is the information functional;
- $\mathcal{E}$ is an environment or discarded algebra;
- $\mathcal{J}$ is an instrument or intervention set;
- $\mathcal{R}$ is the accessible measurement record.

The historical legacy formula supplies none of the last six objects uniquely. **[Proven]**

Chapter 3

Part II — Program 31: loss-only legacy-to-strict bridge

3.1 Question

Can the missing completion be supplied solely by a negative-information retention factor?

3.2 Loss-only ansatz

For $d = 1, \dots, 6$, suppose

$$K_s(d) = c\, r_d K_\ell(d), \qquad c \neq 0, \qquad 0 < r_d \leq 1.$$

The factor r_d may remove amplitude but cannot rotate phase or mix distances.

3.3 Exact sign obstruction

The computed rows are:

d	$K_\ell(d)$	$K_s(d)$	K_s/K_ℓ
1	$+0.710493827$	$+0.469985673$	$+0.661491563$
2	-1.359112119	$+0.192043552$	-0.141300743
3	-2.600111701	$+0.091428614$	-0.035163341
4	-2.308781027	$+0.047029169$	-0.020369696
5	-0.683427396	$+0.024131223$	-0.035309125
6	$+1.307824869$	$+0.011070817$	$+0.008465061$

For $c > 0$, four signs disagree. For $c < 0$, the signs at $d = 1, 6$ disagree. Allowing $r_d = 0$ cannot produce the nonzero strict target at a mismatched distance. Therefore no loss-only diagonal multiplier exists. **[Proven]**

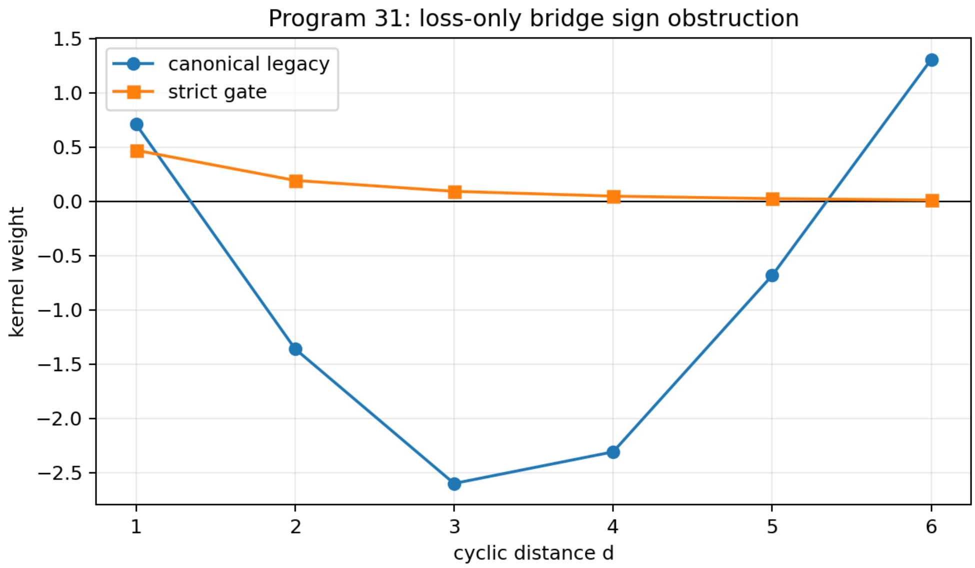

Figure 3.1: Canonical legacy and strict radial rows. Positive attenuation preserves signs and therefore cannot supply the missing phase transformation.

Even with an independently chosen retention for every distance, the positive-normalization residual is bounded below by

$$0.219167,$$

or (42.25%) of the strict-row norm.

3.4 Additive Loewner test

An alternative passive-loss proposal is

$$A_s = A_\ell + B, \qquad B \succeq 0.$$

On C_{12}, the computed difference satisfies

$$\lambda(B) \subset [8.9 \times 10^{-16}, 22.9541],$$

with rank eleven. On a twelve-point interval obtained by extrapolating the same analytic formulas to $|i - j| \leq 11$,

$$\lambda(B) \subset [-4.69964, 20.1323].$$

Thus the additive dissipative interpretation works on the declared cycle realization but is not support-independent. **[Strong numerical evidence]**

3.5 Verdict

- Complete bridge by positive attenuation alone: **[Refuted]**.

- Passive additive correction on C_{12}: **[Strong evidence]**.
- Geometry-independent completion theorem: [**Refuted by the interval counterexample**].
- Required additional ingredient: a phase/frequency transformation or non-diagonal distance-mixing map. **[Proven]**

Chapter 4

Part III — Program 32: Markov admissibility of the legacy kernel

4.1 Question

Does the canonical signed legacy kernel itself generate classical information loss?

4.2 Candidate generator

Define the circulant zero-diagonal matrix W_ℓ on C_{12} and

$$A_\ell = \mathrm{diag}(W_\ell \mathbf{1}) - W_\ell.$$

A continuous-time Markov generator $-A_\ell$ requires nonnegative off-diagonal transition rates, equivalently $(W_\ell)_{xy} \geq 0$ for $x \neq y$.

4.3 Failure of the raw signed row

The row sum is

$$s_\ell = -11.174051962,$$

and

$$\sigma(A_\ell) \subset [-21.377059810, 0].$$

The matrix exponential already violates positivity at short time:

t	$mine^{-tA_\ell}$	$\|e^{-tA_\ell}\|_2$
0.001	-0.00263292	1.02161
0.01	-0.0295213	1.23834
0.1	-1.06641	8.47996

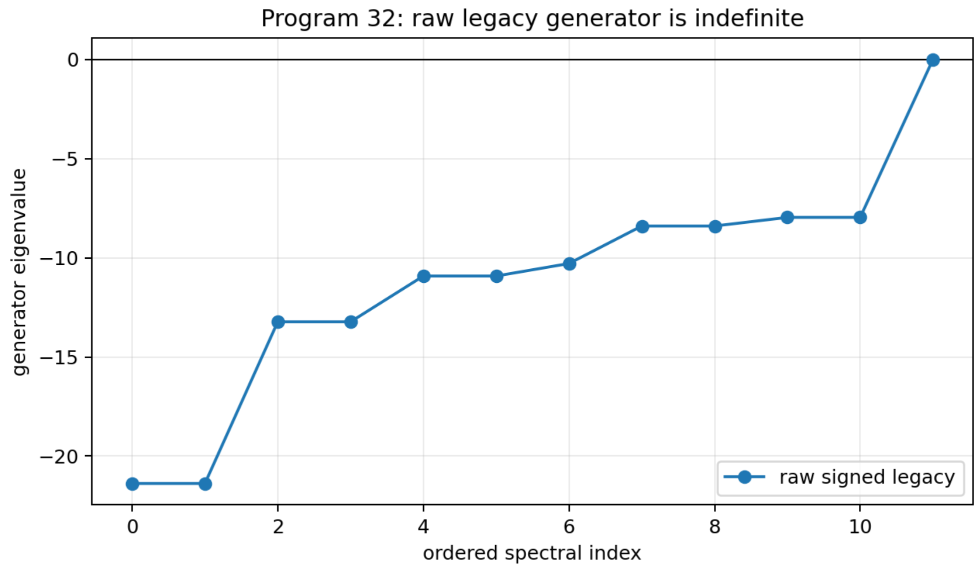

Figure 4.1: Spectrum of the raw signed legacy Laplacian candidate on the twelve-cycle. Eleven nonconstant eigenvalues are negative.

The raw legacy row is not a stochastic information-loss generator. **[Proven]**

4.4 Nonunique positivity repairs

Two elementary replacements are

$$W_\ell^+ = \max(W_\ell, 0), \qquad W_\ell^{|\cdot|} = |W_\ell|.$$

Both yield symmetric Markov semigroups, but:

Repair	**Spectral gap**	$\mathcal{I}(P_1\delta_0)$
positive part	0.710493827	0.261871502
absolute value	15.185443105	7.73×10^{-14}

The gap ratio is 21.3731. The same static signed row therefore admits radically different loss dynamics after equally elementary nonlinear repairs. **[Proven]**

4.5 Verdict

- Raw legacy kernel as a Markov loss law: **[Refuted]**.
- Existence of repaired Markov laws: **[Proven]**.
- Unique repair selected by the legacy formula: **[Refuted]**.
- A microscopic or operational repair criterion remains necessary.

Chapter 5

Part IV — Program 33: hazard reconstruction of nonlinear strict damping

5.1 Envelope-only question

After explicitly setting the phase obstruction aside, can the denominator change be written as cumulative negative feedback?

Define

$$E_\ell(d) = \frac{1}{1 + 0.01d}, \qquad E_s(d) = \frac{1}{1 + d^{1.8}},$$

and

$$R_d = \frac{E_s(d)}{E_\ell(d)} = \frac{1 + 0.01d}{1 + d^{1.8}}.$$

5.2 Discrete hazard

Write

$$R_d = \prod_{j=1}^{d} \rho_j, \qquad h_j = -\log \rho_j = \log \frac{R_{j-1}}{R_j}, \qquad R_0 = 1.$$

All twelve hazards are positive. Selected values are:

d	R_d	h_d
1	0.505000	0.683197
2	0.227567	0.797115
3	0.125233	0.597268
6	0.0405233	0.303959
12	0.0126404	0.145740

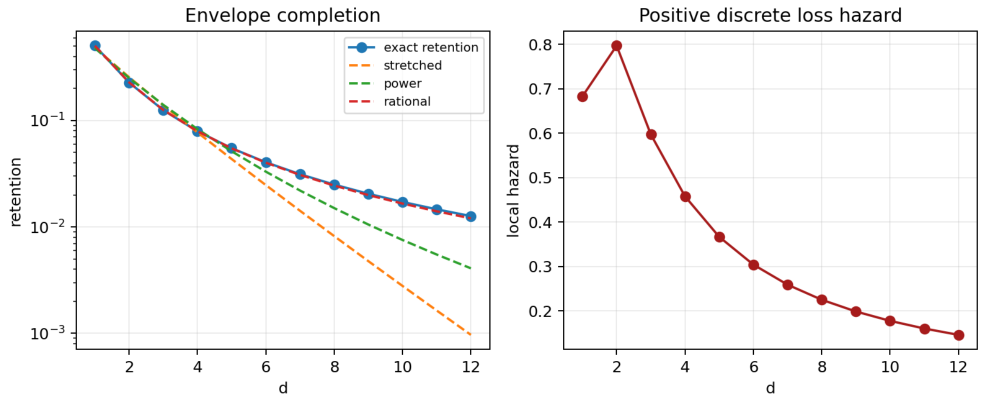

Figure 5.1: Exact legacy-to-strict envelope retention, fitted candidate laws, and the positive discrete hazard sequence.

Hence an exact cumulative-loss representation exists for the envelopes. **[Proven]**

5.3 Competing feedback laws

Four two-parameter or simpler families were fitted without inserting the exact numerator:

Law	**Best parameters**	**Relative L^2 error**	**Maximum relative error**
$e^{-\gamma d}$	$\gamma = 0.687293$	0.10797	0.9793
$e^{-\gamma d^q}$	$\gamma = 0.727191, q = 0.907984$	0.09705	0.9236
$(1 + cd)^{-p}$	$c = 0.142184, p = 5.52676$	0.07769	0.6774
$(1 + bd^\eta)^{-1}$	$b = 0.982316, \eta = 1.78137$	0.003233	0.0487

Under (1%) multiplicative perturbations, the rational-fit (95%) interval was approximately

$$\eta \in [1.7506, 1.8125].$$

This supports the numerical robustness of the exponent within the declared finite family. It does not source the exponent. **[Strong evidence]**

5.4 Nonuniqueness theorem

Every strictly decreasing positive sequence admits a unique discrete hazard sequence. The same hazard can be implemented by erasure, dephasing, hidden-state mixing, environmental leakage, absorption, or feedback. Therefore the hazard representation does not identify a microscopic mechanism. **[Proven]**

5.5 Verdict

- Positive envelope-loss representation: **[Proven]**.
- Unique microscopic negative coupling: **[Refuted]**.
- Derivation of $\eta = 1.8$ from information theory: [**Not established**].
- Phase completion: **still absent by Program 31**.

Chapter 6

Part V — Program 34: stability cone of negative feedback

6.1 Two placements of feedback

Using the positive strict weights w_d, compare

$$w_d^{\exp}(\gamma) = w_d e^{-\gamma d}$$

with

$$w_d^{\text{lin}}(\gamma) = w_d(1 - \gamma d).$$

Both express decreasing coupling, but only the first is automatically positive.

6.2 Exact admissibility boundary

Because $w_d > 0$ for $d = 1, \ldots, 6$, the linear law has nonnegative rates precisely when

$$0 \leq \gamma \leq \frac{1}{d_{\max}} = \frac{1}{6}.$$

Beyond this value, at least one off-diagonal transition rate is negative. **[Proven]**

Selected results are:

Law	γ	**Minimum weight**	**Minimum eigen-value**	**Gap**
exponential	0.4	+0.00100432	numerical zero	0.268463
linear	1/6	0	numerical zero	0.386440
linear	0.2	−0.00221416	numerical zero	0.312904
linear	0.4	−0.0282175	−0.128313	—

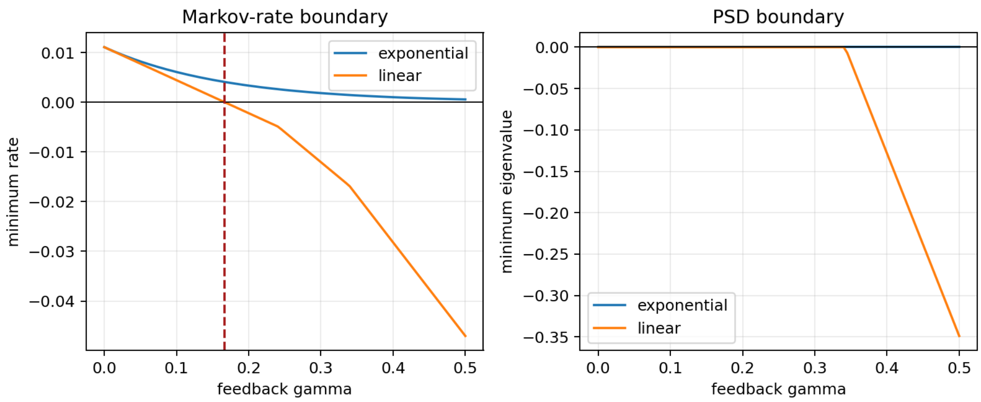

Figure 6.1: Markov-rate and spectral boundaries for exponential retention and linear subtraction.

At $\gamma = 0.2$,

$$\min e^{-0.001A_\gamma} = -2.21 \times 10^{-6},$$

so Markov positivity fails immediately even though the quadratic form has not yet become negative. At $\gamma = 0.4$, both rate and spectral conditions fail.

One hundred random-distribution tests found no relative-information increase for exponential $\gamma = 0.4$ or boundary-linear $\gamma = 1/6$. [**Strong numerical evidence consistent with data processing**]

6.3 Verdict

- Exponential retention: [**Proven Markov-admissible**].
- Unrestricted linear subtraction: [**Refuted**].
- Positive semidefiniteness alone as a stochastic criterion: [**Refuted**].
- Physical choice of γ: **not supplied**.

Chapter 7

Part VI — Program 35: information destruction versus environmental transfer

7.1 Declared channel and dilation

For the strict stochastic semigroup $P_t = e^{-tA_s}$, define the measure-and-prepare channel

$$\Phi_t(\rho) = \sum_{x,y} P_t(y|x)\langle x|\rho|x\rangle|y\rangle\langle y|.$$

It has the isometry

$$V_t|x\rangle = \sum_y \sqrt{P_t(y|x)}|y\rangle_S|x,y\rangle_E.$$

The finite computation gives

$$\|V_t^\dagger V_t - I\|_F = 1.06 \times 10^{-15}$$

and a reduced-channel residual 1.39×10^{-16}. This is a direct instance of Stinespring dilation. **[Proven]**

7.2 Information ledger

Starting from δ_0, at $t = 1$:

$$H(p_1) = 2.225996234, \qquad D(p_1\|u) = 0.258910416,$$

$$\operatorname{Tr}\rho_S^2 = 0.135124653.$$

For the pure dilation,

$$S(SE) = 0, \qquad S(S) = S(E) = 2.225996234,$$

and

$$I(S : E) = 4.451992468.$$

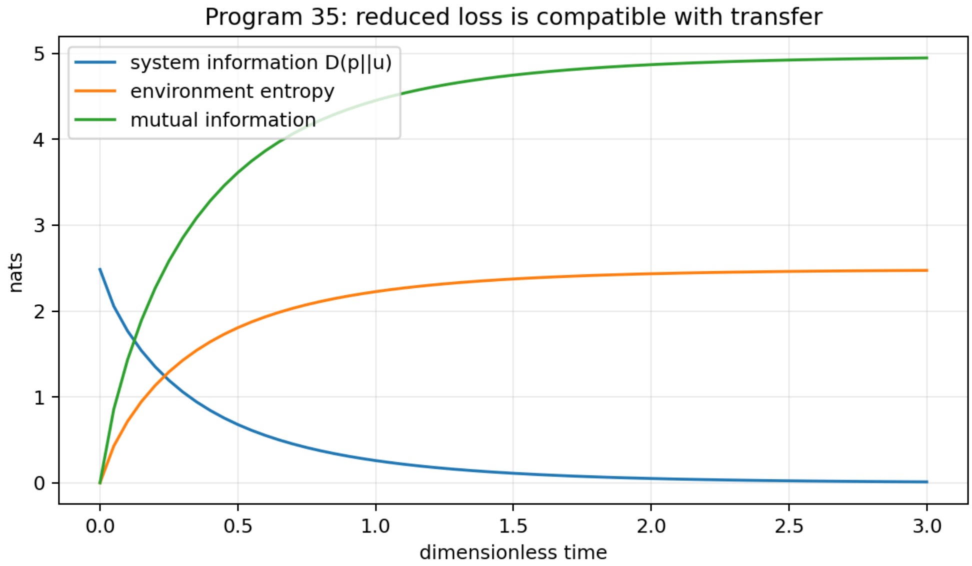

Figure 7.1: Reduced-system information, environment entropy, and mutual information for the explicit isometric dilation.

The reduced system loses accessible information while the joint description remains pure. This is compatible with the general principle that reduced open-system loss need not be fundamental deletion. **[Proven]**

7.3 Identifiability obstruction

Every finite completely positive trace-preserving channel admits an isometric dilation. Therefore system-only observations cannot decide whether their reduced loss is fundamental or environmental. One needs environment access, recovery operations, or an additional axiom declaring the system closed. **[Proven]**

The no-hiding theorem further constrains where quantum information can reside in a unitary embedding, but the present result needs only Stinespring dilation.

7.4 Verdict

- Reduced-system information loss: **[Proven]**.
- Compatibility with transfer and correlation: **[Proven]**.
- Fundamental destruction inferred from system data: **[Refuted as an inference]**.
- Thermodynamic heat cost: **not derived**.

Chapter 8

Part VII — Program 36: open-system coupling-sign gauge

8.1 Model

Let

$$B = \frac{A_s - A_\ell}{\|A_s - A_\ell\|_F}$$

on the declared cycle and define

$$H_{\mathrm{tot}}(g) = A_s \otimes I_E + I_S \otimes H_E + gB \otimes X_E.$$

The numerical study used a two-level environment,

$$H_E = 0.7\sigma_z, \qquad X_E = \sigma_x, \qquad |g| = 0.8.$$

8.2 Exact parity theorem

Suppose a bath parity P_E satisfies

$$P_E H_E P_E = H_E, \qquad P_E X_E P_E = -X_E, \qquad P_E \tau_E P_E = \tau_E.$$

Then

$$H_{\mathrm{tot}}(-g) = (I \otimes P_E) H_{\mathrm{tot}}(g) (I \otimes P_E),$$

and, after tracing out the bath,

$$\Phi_t^{(-g)} = \Phi_t^{(g)}.$$

Thus the system cannot identify the sign. **[Proven]**

8.3 Numerical sign tests

Across three system probes and 41 times:

$$\max_t D\left(\Phi_t^{(+g)}, \Phi_t^{(-g)}\right) = 0$$

for the symmetric bath.

For a bath polarized along σ_x, the maximum system trace distance became

$$0.0468978.$$

A calibrated odd environment record reached a difference

$$0.368994.$$

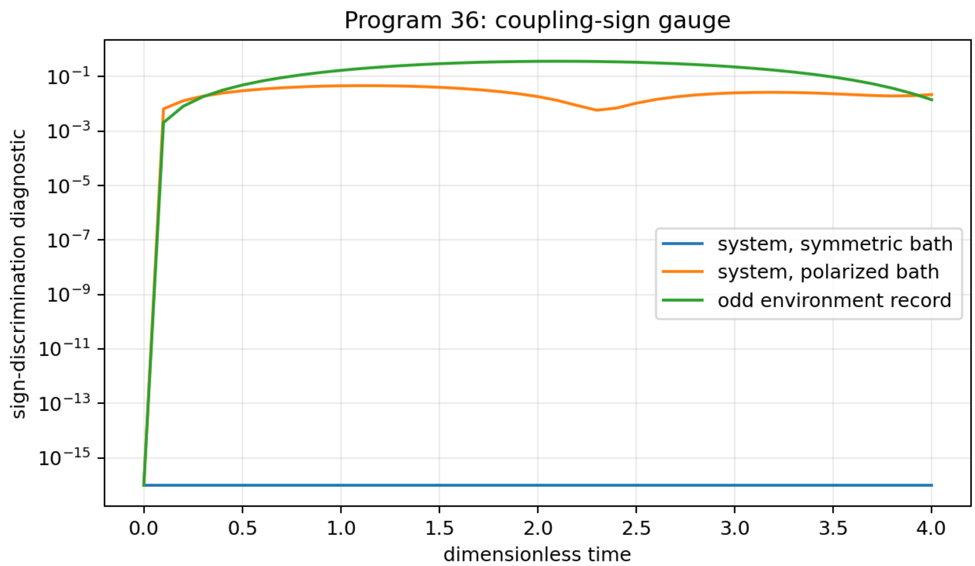

Figure 8.1: System and environment sign-discrimination diagnostics. Symmetric-bath system dynamics is exactly sign blind.

The transformation

$$(g, X_E) \longmapsto (-g, -X_E)$$

is a sign gauge. A statement that $g < 0$ has operational content only after the polarity of X_E is fixed independently. **[Proven]**

8.4 Verdict

- System-only coupling sign in a parity-symmetric environment: [**Unidentifiable**].
- Sign after a calibrated odd reference: [**Identifiable in the declared model**].
- Internal FIN source of that reference: [**Not established**].
- Selector closure from coupling sign: [**Not obtained**].

Chapter 9

Part VIII — Program 37: feedback and entropy-production ledger

9.1 Positive-rate feedback law

Let $b_i = \cos(2\pi i/12)$. Starting from the strict positive rates, define

$$w_{ij}(\theta) = w_{ij}^s \exp\left[-\frac{\theta}{2}(b_i - b_j)\right].$$

These rates satisfy detailed balance with

$$\pi_i(\theta) \propto e^{-\theta b_i}.$$

The controller used

$$\dot{\theta} = -0.8\theta, \qquad \theta(0) = 1.4.$$

The quadratic controller loss fell from 0.98 to 6.64×10^{-5}. The instantaneous Markov entropy-production rate

$$\sigma = \sum_{i<j}(w_{ij}p_j - w_{ji}p_i)\log\frac{w_{ij}p_j}{w_{ji}p_i}$$

remained nonnegative to numerical precision. [**Strong evidence; nonnegativity follows termwise**]

9.2 Stable does not imply variational

Consider the stable two-coordinate controller

$$\dot{\theta} = J\theta, \qquad J = \begin{pmatrix} -0.8 & 1.7 \\ -1.7 & -0.8 \end{pmatrix}.$$

Its eigenvalues are

$$-0.8 \pm 1.7i,$$

so it is asymptotically stable. However,

$$\frac{\|J - J^T\|_F}{\|J\|_F} = 1.80964,$$

and the work around the unit circle is

$$\oint F \cdot d\theta = -10.6814 \neq 0.$$

No scalar potential generates this feedback in the declared Euclidean coordinates. **[Proven]**

9.3 Thermodynamic accounting boundary

Feedback thermodynamics requires measurement records, controller memory, work, and mutual-information terms, as made explicit by the Sagawa–Ueda feedback relation. A decreasing system functional alone is not a complete entropy ledger.

9.4 Verdict

- Stable negative feedback: [**Constructed**].
- Compatibility with nonnegative Markov entropy production: [**Proven for the model**].
- Automatic existence of an action: **[Refuted]**.
- Kernel-derived controller, target, or polarity: [**Not established**].

Chapter 10

Part IX — Program 38: wave–diffusion observer test

10.1 Common positive feedback family

Define

$$w_d(g) = w_d^s e^{gd/6}, \qquad A(g) = D(g) - W(g).$$

The same signed parameter appears in

$$U_t(g) = e^{-itA(g)}$$

and

$$P_t(g) = e^{-tA(g)}.$$

No positivity problem arises for finite g.

10.2 Short-time theorem

For a localized preparation $|x\rangle$,

$$P_t(x|x) = 1 - A_{xx}t + O(t^2),$$

whereas

$$|U_t(x|x)|^2 = 1 - \mathrm{Var}_x(A)t^2 + O(t^4).$$

The fitted escape exponents were:

g	**diffusion**	**unitary wave**
-0.5	0.99907	2.00000

g	**diffusion**	**unitary wave**
0	0.99895	2.00000
+0.5	0.99879	2.00000

No clock rescaling changes a generic linear leading term into a quadratic one. **[Proven]**

10.3 Chapman–Kolmogorov and phase records

For diffusion,

$$P_{2t} = P_t^2$$

to residuals below 4×10^{-15}.

For unitary population matrices

$$M_t(y|x) = |U_t(y,x)|^2,$$

the Chapman–Kolmogorov residual reached 0.93, because unobserved amplitudes interfere.

For the double-path input

$$|\psi_\varphi\rangle = \frac{|0\rangle + e^{i\varphi}|3\rangle}{\sqrt{2}},$$

the best-detector visibility remained between 0.966 and 0.9995 over the tested g-values.

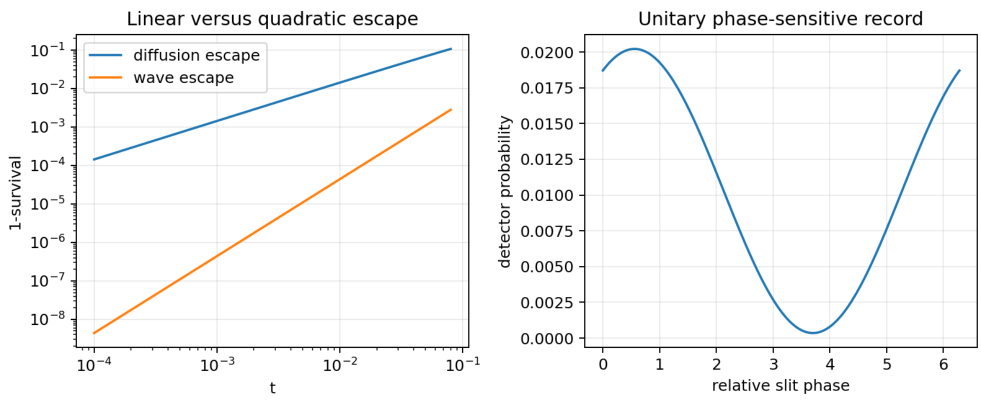

Figure 10.1: Linear diffusive versus quadratic unitary escape and a phase-sensitive double-path record.

Intermediate interventions and phase variation therefore distinguish the temporal categories without invoking an undefined observer collapse. This is consistent with the operational process-tensor approach of Pollock et al..

10.4 Verdict

- Common spatial feedback parameter: [**Constructed**].
- Selection of wave versus diffusion by its sign: **[Refuted]**.
- Operational distinction by early time, interventions, or phase: **[Proven]**.
- Physical clock or observer ontology: [**Not derived**].

Chapter 11

Part X — Program 39: structural identifiability of sign and scale

11.1 Gauge-blind observation model

The finite identification model uses Poisson means

$$\lambda_r = \lambda_0 + \epsilon + E_r\big[\tau(gb_r + hc_r)\big]^2,$$

where g is the signed coupling, tau is clock scale, epsilon is background, and h c_r is an optional calibrated reference.

Without the reference h=0, only the product tau g is observed and the likelihood is symmetric under sign reversal.

11.2 Fisher-rank test

For the declared true point

$$(g, \tau, \epsilon) = (-0.42, 1.06, 1.3),$$

the no-reference Fisher eigenvalues were

$$(1.54 \times 10^{-14}, 0.2310, 417.626),$$

so the rank was two.

With a calibrated reference they became

$$(0.1078, 113.666, 615.176),$$

and the rank became three. [**Strong numerical evidence**]

The no-reference sign difference in profile negative log likelihood was exactly zero. With the reference it was

$$\Delta\mathrm{NLL} = 24.3252$$

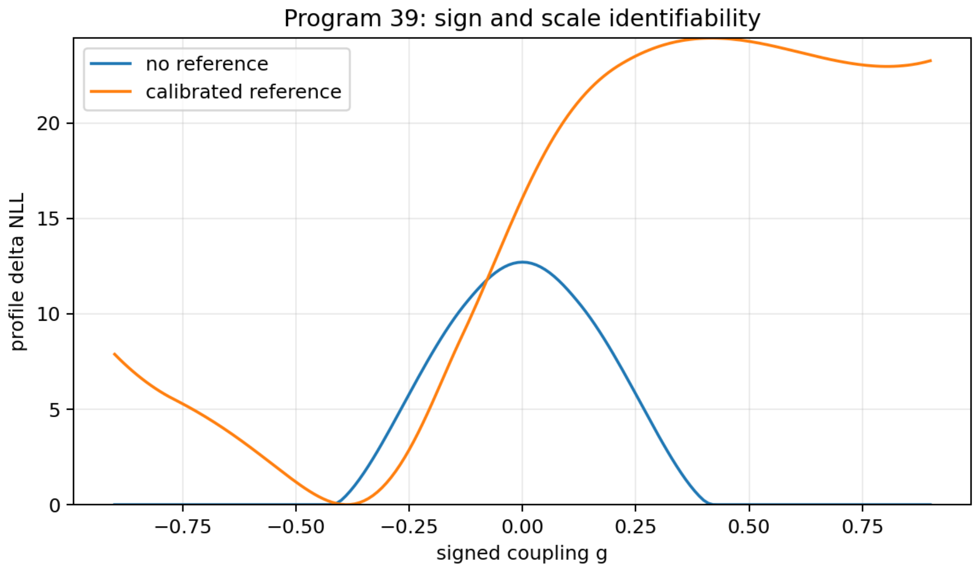

Figure 11.1: Profile likelihood for the signed coupling with and without a calibrated reference channel.

at the true magnitude. The fitted sign was $g = -0.39$. Reversing apparatus polarity without relational recoding changed the fit to $g = +0.39$.

11.3 Identifiability theorem for the declared family

The transformations

$$(g, B) \sim (cg, B/c), \qquad (A, \tau) \sim (cA, \tau/c), \qquad (g, X_E) \sim (-g, -X_E)$$

are observational equivalences unless an amplitude, clock, and polarity reference is supplied. A nonzero signed coupling is not an invariant of uncalibrated data. **[Proven]**

11.4 Verdict

- Sign without reference: [**Globally unidentifiable**].
- Sign with independently calibrated interference: [**Identifiable in the model**].
- Strict internal source of clock, amplitude, and polarity: [**Absent**].
- A prior preference for $g < 0$ as evidence: [**Rejected**].

Chapter 12

Part XI — Program 40: blinded negative-coupling challenge

12.1 Frozen synthetic protocol

Four models were declared before generating the counts:

- negative operational coupling, $g = -0.28$, with a correlated odd environment record;
- zero coupling;
- positive operational coupling, $g = +0.28$;
- a static mimic with the same system operator as the negative model but no correlated environment record.

Training used diffusion at $t = 0.35$, unitary populations at $t = 0.55$, and one environment record. Held-out data used diffusion at $t = 1.05$, unitary populations at $t = 1.35$, a double-path record at $t = 1.8$, and two environment records. Clock and SPAM nuisance parameters were fitted on training data only.

The protocol commitment was

```
1778a5f93f824bc09a4948c9124b28bb23e823b950970a1b01c51aa3bcd1903b
```

This is a deterministic internal commitment, not an externally timestamped preregistration.

12.2 Results

With 500 shots per block and 60 trials per model, the confusion matrix was

$$\begin{pmatrix} 56 & 4 & 0 & 0 \\ 0 & 49 & 1 & 10 \\ 0 & 1 & 58 & 1 \\ 0 & 18 & 0 & 42 \end{pmatrix},$$

in the order negative, zero, positive, static mimic.

Overall accuracy was

$$0.85417.$$

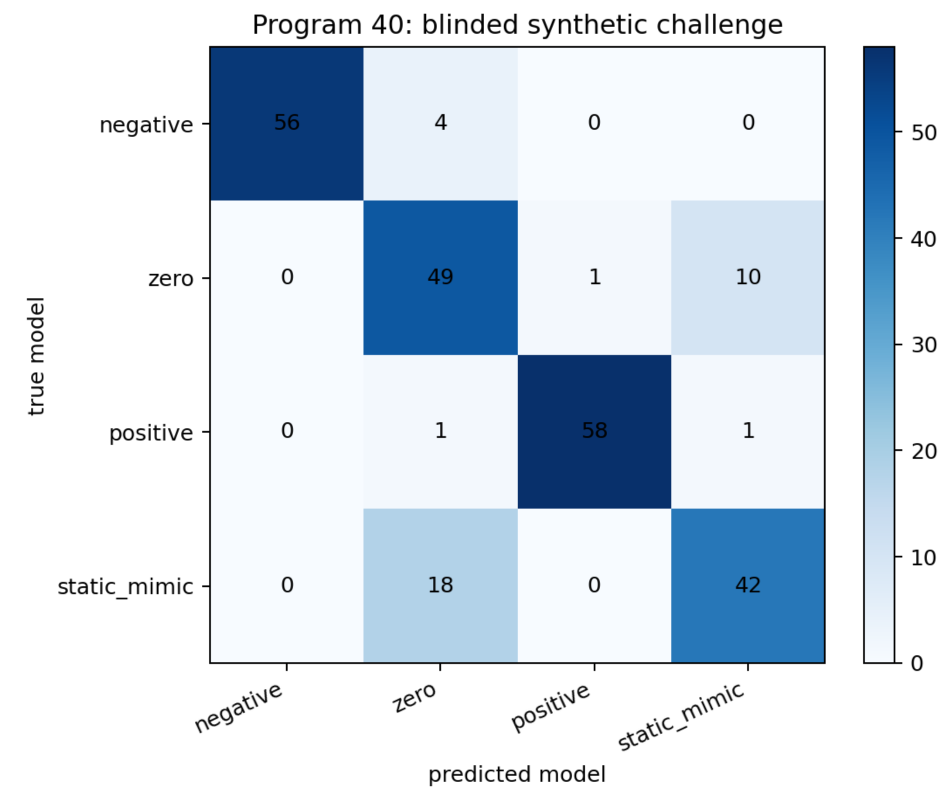

Figure 12.1: Confusion matrix for the four-model blinded synthetic held-out challenge.

Per-model accuracies were:

Model	Accuracy
negative	0.9333
zero	0.8167
positive	0.9667
static mimic	0.7000

The median winning held-out log-likelihood margin was 5.49464.

12.3 Causal nonidentifiability result

After removing environment records, the negative operational model and the static mimic had system-signature residual

$$0.$$

They are exactly the same system process by construction. No amount of additional system-only data can distinguish "dynamical negative coupling" from "the same static operator" in this candidate pair. The environment record, intervention on the coupling, or a different-size prediction is essential. **[Proven]**

12.4 Verdict

- Synthetic pipeline sensitivity to the negative model: **[Strong evidence]**.
- Unique causal identification from system dynamics: **[Refuted]**.
- Hardware validation: [**Not executed**].
- Strict legacy-to-strict source theorem: [**Not obtained**].
- Evidence for fundamental physics: **none**.

Chapter 13

Part XII — Can negative information coupling be placed in an action?

13.1 Static quadratic correction

On C_{12}, the positive difference $B = A_s - A_\ell \succeq 0$ permits the Euclidean quadratic correction

$$S_s[\phi] = S_\ell[\phi] + \frac{1}{2}\phi^T B\phi.$$

This stabilizes a static precision form. It does not prove temporal dissipation or information loss: the same positive operator can enter either e^{-itA} or e^{-tA}. **[Proven]**

13.2 Why an ordinary potential is insufficient

A conservative action

$$S[q] = \int L(q, \dot{q})\, dt$$

does not by itself encode irreversible loss. Classical damping may be represented conditionally by a Rayleigh function

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} + \frac{\partial \mathcal{R}}{\partial \dot{q}} = 0,$$

or statistically by an Onsager–Machlup functional. These are additional dynamical structures.

Program 37 gives an explicit stable feedback field with nonzero circulation. It cannot be obtained as the gradient of a scalar potential in the declared coordinates. A post-hoc (V(K)) therefore does not solve the operational loss problem.

13.3 Minimal closed-system action with an environment

A standard conditional construction is

$$S[\phi, q_E] = S_s[\phi] + S_E[q_E] + g \int dt\, \phi^T B q_E.$$

The total theory may remain unitary or Hamiltonian. Eliminating q_E produces a nonlocal influence functional, noise, and dissipation for the reduced field. This is the logic of the Feynman–Vernon influence functional and Caldeira–Leggett model.

The sign-gauge theorem persists: g has no absolute meaning unless the environment coordinate has calibrated polarity.

13.4 Relation to inverse kernel reconstruction

The covariance-level functional

$$\Gamma_Q(G) = \frac{1}{2}\,\mathrm{Tr}(QG) - \frac{1}{2}\log\det G$$

still reconstructs $G = Q^{-1}$ conditionally. It does not derive the dissipative coefficient, bath spectrum, information functional, or measurement trace. The inverse-propagator problem and the open-system loss problem are logically independent. **[Proven]**

13.5 Action verdict

The smallest mathematically honest action-level interpretation is:

$$\boxed{\text{strict precision} + \text{environment action} + \text{explicit interaction} + \text{declared coarse graining}}$$

not a single new scalar term in (V(K)).

Chapter 14

Part XIII — Integrated theorem and obstruction inventory

14.1 Newly established results

14.1.1 Theorem A — loss-only phase no-go

A positive distancewise retention cannot transform the canonical legacy row into the strict row because their sign patterns differ. **[Proven]**

14.1.2 Theorem B — signed-kernel Markov obstruction

The canonical legacy row has negative off-diagonal weights and therefore does not define a continuous-time Markov generator. **[Proven]**

14.1.3 Theorem C — hazard existence and microscopic nonuniqueness

Every positive decreasing envelope has a discrete nonnegative hazard, but the hazard does not identify its microscopic realization. **[Proven]**

14.1.4 Theorem D — linear-feedback boundary

For the strict six-distance row, $w_d(1-\gamma d)$ is Markov-admissible exactly for $0 \le \gamma \le 1/6$. **[Proven]**

14.1.5 Theorem E — reduced-loss dilation obstruction

System-only data cannot distinguish a finite CPTP loss channel from its isometric environment realization. **[Proven]**

14.1.6 Theorem F — coupling-sign parity gauge

Under a parity-symmetric bath, reduced dynamics is invariant under $g \mapsto -g$. **[Proven]**

14.1.7 Theorem G — common-feedback temporal nonselection

The same positive feedback-modified spatial generator supports both unitary and Markov evolution; its sign does not choose the temporal category. **[Proven]**

14.1.8 Theorem H — static-mimic causal nonidentifiability

Two hypotheses that induce the same complete system process cannot be causally distinguished without environment or intervention data. **[Proven]**

14.2 Refuted proposals

- Negative entries of K_ℓ are information loss. **[Refuted]**
- The historical attenuation diagram is already an operational loss theorem. **[Refuted]**
- Positive retention alone completes legacy into strict. **[Refuted]**
- The raw legacy row is a Markov generator. **[Refuted]**
- A positive quadratic correction uniquely selects diffusion rather than waves. **[Refuted]**
- Stable negative feedback must be a gradient flow. **[Refuted]**
- A signed coupling is observable without a polarity reference. **[Refuted]**
- Reduced entropy increase proves fundamental information destruction. **[Refuted]**

14.3 Surviving conditional bridge

The smallest bridge compatible with all falsification tests has at least

$$K_\ell \xrightarrow{\mathcal{A}} K_{\text{normalized}} \xrightarrow{\mathcal{P}} K_{\text{phase reorganized}} \xrightarrow{\mathcal{R}_{\text{loss}}} K_s,$$

where:

- $\mathcal{A}$ absorbs or reinterprets α_{geo};
- $\mathcal{P}$ changes phase/frequency or mixes distance sectors;
- $\mathcal{R}_{\text{loss}}$ supplies a positive nonlinear compression envelope.

For operational physics this must be augmented by

$$(\mathcal{S}, \mathcal{T}_t, \mathcal{I}, \mathcal{E}, \mathcal{J}, \mathcal{R}_{\text{meas}}).$$

No calculation here sources $\mathcal{A}$, $\mathcal{P}$, the feedback rate, the clock, the environment, or the apparatus from the strict core. This is a minimal candidate architecture, not a closure theorem.

Chapter 15

Part XIV — Relation to existing mathematics and physics

FIN question	Established structure	Similarity	Essential difference
positive loss semigroup	Markov generators and data processing	relative information contracts	legacy signed weights are not rates
local quantum loss	GKSL/Lindblad semigroups	completely positive dephasing and relaxation	bath and rates are new inputs
hidden environment	Stinespring dilation	reduced loss from global isometry	does not select a physical bath
feedback information	stochastic thermodynamics	controller information enters entropy ledger	FIN supplies no controller or temperature
nonconservative action	Onsager–Machlup/ influence functional	dissipation uses paths or doubled/ environment variables	not a potential of (K) alone
sign identification	system identification with reference interference	odd cross terms reveal sign	reference polarity is an axiom/calibration
wave–diffusion ambiguity	quantum walks versus Markov semigroups	same graph operator supports both	temporal category remains independent

The main primary references are:

- Stinespring, positive functions on C^*-algebras.
- Gorini, Kossakowski, and Sudarshan, completely positive dynamical semigroups.
- Lindblad, generators of quantum dynamical semigroups.
- Breuer, Laine, and Piilo, information-backflow criterion.
- Sagawa and Ueda, feedback fluctuation relation.
- Reeb and Wolf, rigorous Landauer principle.
- Pollock et al., operational Markov condition and process tensors.
- Feynman and Vernon, influence functionals.
- Caldeira and Leggett, dissipative quantum systems.

The proposed negative-information extension is therefore not a new mathematical category. Its possible FIN-specific content would have to lie in an independently derived bridge map, environment coupling, or predictive restriction—not in the generic existence of dissipation or feedback.

Chapter 16

Part XV — Information-to-thermodynamics boundary

The dimensionless entropy change in Program 35 does not determine heat. A Landauer statement requires at least a reservoir Hamiltonian and an inverse temperature. In a finite quantum setup, the exact entropy–heat ledger also includes correlations and reservoir relative entropy, as shown by Reeb and Wolf.

Under the scaling

$$H_E \mapsto cH_E, \qquad \beta \mapsto \beta/c,$$

the Gibbs state $e^{-\beta H_E}/Z$ is unchanged while the dimensional energy scale changes. A dimensionless channel cannot select c. Therefore neither $k_B T$, heat, nor an energy unit follows from the computed information loss. **[Proven]**

The minimal physical bridge must explicitly supply:

1. an energy or action unit;
2. a physical clock;
3. a bath Hamiltonian;
4. a preparation state such as a Gibbs state;
5. a coupling calibration;
6. an apparatus record.

These are conversion and operational axioms unless independently sourced.

Chapter 17

Part XVI — Recommended Programs 41–50

The following studies are ranked by probability of delivering a technically decisive result, not by probability of confirming FIN.

1. **Program 41 — support-independent Loewner bridge theorem.** Classify graph supports and normalization conventions for which $A_s - A_\ell \succeq 0$. Success probability: 0.80.
2. **Program 42 — minimal phase-map reconstruction.** Solve for the sparsest equivariant distance-mixing operator $\mathcal{P}$ that converts the legacy carrier phase to the strict carrier before attenuation. Success probability: 0.65.
3. **Program 43 — held-out-size hazard law.** Fit the loss hazard only on C_{12}, then test $N = 16, 24, 48$ completions fixed in advance. Success probability: 0.75.
4. **Program 44 — CP-divisibility and information-backflow audit.** Compare Markovian dephasing, finite memory, and collision baths under the same strict operator. Success probability: 0.85.
5. **Program 45 — environment recovery experiment.** Implement the Program 35 dilation and test whether the apparently lost record can be recovered from the ancilla. Success probability: 0.90 on an emulator.
6. **Program 46 — calibrated sign-reference emulator.** Build the (gB+hC) interference design and preregister sign, clock, and SPAM confidence criteria. Success probability: 0.70.
7. **Program 47 — influence-functional reconstruction.** Determine the smallest bath spectral density whose reduced kernel produces the fitted strict hazard without post-hoc target insertion. Success probability: 0.40.
8. **Program 48 — full feedback thermodynamic ledger.** Measure controller work, mutual information, bath flow, and coarse-grained entropy production in one closed accounting scheme. Success probability: 0.60.
9. **Program 49 — process-tensor causal challenge.** Distinguish operational negative coupling from a static operator mimic using interventions rather than final-time fits. Success probability: 0.65.
10. **Program 50 — externally preregistered multi-size challenge.** Freeze bridge maps and alternative models before receiving data from two graph sizes. Success probability: 0.50.

Stop rules:

- Do not call a signed matrix entry information loss.
- Do not infer a Markov process from positive semidefiniteness alone.
- Do not use rectification $K \mapsto |K|$ without declaring it as a new nonlinear map.
- Do not call an exact envelope factorization a microscopic derivation.
- Do not infer coupling sign without a calibrated odd reference.

- Do not infer fundamental deletion from a reduced channel.
- Do not infer a thermodynamic cost without temperature and energy calibration.
- Do not promote a C_{12}-only bridge to a support-independent theorem.
- Do not transfer legacy physical roles to strict before bridge completion and role audit.
- Do not call synthetic classification an experimental validation.

Chapter 18

Final scientific judgment

The negative-information hypothesis survives, but only after being sharply reformulated.

It does not survive as the assertion that negative legacy entries destroy information. It does not survive as a loss-only derivation of the strict kernel. It does not survive as a unique Markov law, an automatic Lagrangian term, a source of physical time, or a fundamental deletion mechanism.

It survives as the following conditional mathematical architecture:

$$\boxed{\begin{array}{l}\text{legacy precursor} \xrightarrow{\text{normalization + phase completion}} \text{positive strict carrier} \\ \xrightarrow{\text{declared retention/dissipator}} \text{reduced information dynamics} \\ \xrightarrow{\text{environment + instrument + record}} \text{operationally testable loss model.}\end{array}}$$

The envelope component is mathematically plausible and exactly representable by positive hazards. The phase component is unavoidable and independent. The operational component requires a state, clock, environment, instrument, and information measure. The sign component requires a calibrated polarity reference. The thermodynamic component requires dimensional conversion axioms.

Thus negative information coupling is a viable new research layer around FIN, but it is neither contained in the legacy kernel nor forced by the strict kernel. Its scientific value will depend on whether Programs 41–50 can turn the current nonunique completion into a source-derived, held-out predictive, and operationally calibrated law.

Chapter 19

Reproducibility statement

The deterministic computation is contained in

fin_programs_31_40_negative_information.py.

It writes

FIN_Programs_31_40_Negative_Information_Results.json

and the nine figures included in this report. The fixed random seed is 20260720. NumPy and SciPy perform all matrix, optimization, integration, and finite-shot calculations.

The synthetic challenge is a software experiment. Its commitment is internal and deterministic, not an externally timestamped preregistration. No hardware data were used.

19.1 Suggested citation before DOI assignment

Żuchowski, K. (2026). *FIN Negative Information Coupling: Executed Programs 31–40, the Legacy-to-Strict Bridge, and the Operational Meaning of Information Loss* (FIN Research Supplement, Release 10.3; Version 1.0.0) [Preprint]. Zenodo.

Archival note

The machine-readable record is `FIN_Programs_31_40_Negative_Information_Results.json`. It is generated by `fin_programs_31_40_negative_information.py`. All calculations are dimensionless and use declared finite models.

Suggested citation before DOI assignment:

Żuchowski, K. (2026). *FIN Negative Information Coupling: Executed Programs 31–40, the Legacy-to-Strict Bridge, and the Operational Meaning of Information Loss* (FIN Research Supplement, Release 10.3; Version 1.0.0) [Preprint]. Zenodo. DOI to be assigned.